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Mobile Communications

Unit 2 Lab Report: Path Loss, Modulation Performance, OFDM and Wireless Network Technologies

Course	Mobile Communications, Mini Projects
Unit	Unit 2 (Friis path loss, Shannon-Hartley capacity, digital modulation, OFDM, WPAN / WLAN / WMAN technologies)
Mini projects	2.1 Free-space path loss and digital modulation performance in Python; 2.2 OFDM waveform generator with a cyclic-prefix visualizer; 2.3 Comparing WPAN, WLAN and WMAN technologies on range, throughput and power
Date	27 October 2026
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1. Introduction

This report documents the three Unit 2 mini projects. Projects 2.1 and 2.2 are pure software simulations in Python with NumPy, SciPy and Matplotlib (no SDR hardware). Project 2.3 is a data-driven comparison built from published specifications. Random seeds are fixed, so every number below can be reproduced by re-running the delivered scripts (`mc_2_1.py`, `mc_2_2.py`, `mc_2_3.py`, `mc_common.py`, `wireless_dataset.csv` and the printed outputs).

- **Project 2.1** implements the Friis equation, verifies the 6 dB-per-doubling behaviour, and measures bit error rate (BER) by Monte Carlo simulation for BPSK, QPSK, 16-QAM and 64-QAM over an additive white Gaussian noise (AWGN) channel, relating the measured gaps to the Shannon-Hartley theorem.
- **Project 2.2** builds a complete OFDM transmitter and receiver, passes the signal through a synthetic multipath channel, and shows how the cyclic prefix (CP) removes inter-symbol interference (ISI).
- **Project 2.3** assembles a sourced dataset for Bluetooth 5, ZigBee, Wi-Fi 6 and WiMAX, visualizes the range, throughput and power trade-off, and recommends a technology for three deployment scenarios.

2. Mini Project 2.1: Free-Space Path Loss and Digital Modulation Performance

2.1 Objective

To build intuition for the Friis free-space path loss equation and the Shannon-Hartley capacity theorem by implementing them in Python, and to measure the BER of BPSK and 16-QAM (and, as extensions, QPSK and 64-QAM) over an AWGN channel, connecting the measured gaps to the theoretical capacity bound.

2.2 Method

- **Friis:** `friis_rx_power` returns P_RX(dBm) = P_TX + G_TX + G_RX − 20·log10(d_km) − 20·log10(f_MHz) − 32.44 element-wise. Link parameters: P_TX = 43 dBm, G_TX = 15 dBi, G_RX = 0 dBi; distances 0.05 to 10 km (500 points); carriers 900, 1900 and 3500 MHz.
- **Mappers:** BPSK maps bit 0 to −1 and bit 1 to +1. QPSK, 16-QAM and 64-QAM are Gray-coded square constellations (levels ±1, ±3, ±5, ±7 as needed) normalized to unit average symbol energy. Demapping is minimum-distance. Unit tests confirm an exact noiseless round trip and unit average energy for every scheme.
- **Channel:** complex AWGN with noise density set from the target Eb/N0 (Es = 1, Eb = Es/k for k bits per symbol). Eb/N0 was swept from 0 to 16 dB in 1 dB steps with 2.4 million bits per scheme.
- **Theory curves:** BPSK and QPSK use 0.5·erfc(√(Eb/N0)); square M-QAM uses the standard approximation (2/k)(1 − 1/√M)·erfc(√(3k·Eb/N0 / (2(M − 1)))).

2.3 Results

Friis path loss

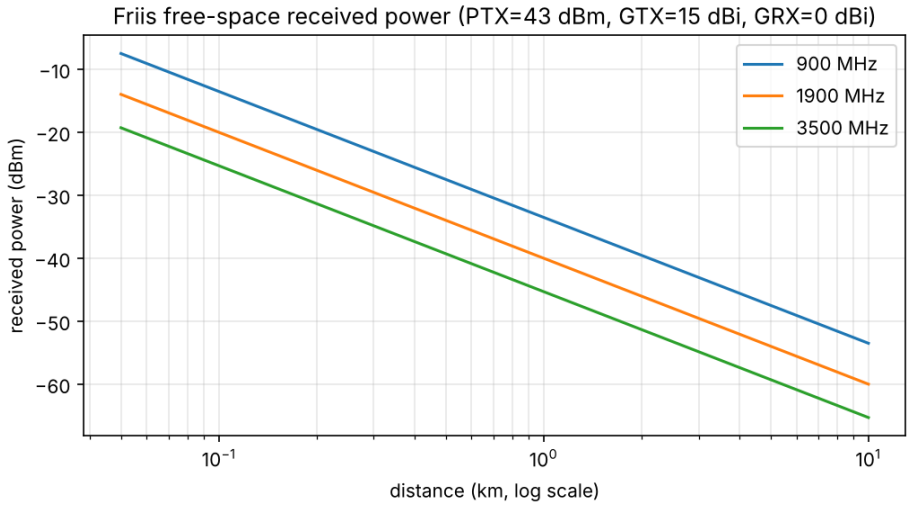


Figure 1: *Received power against distance at 900, 1900 and 3500 MHz (log distance axis).*

On the logarithmic distance axis each curve is a straight line, and the higher the carrier frequency the lower the curve at every distance. The 6 dB-per-doubling relationship was verified numerically by

subtracting the received powers at two distances a factor of two apart, and likewise at two frequencies a factor of two apart (Table 1).

Table 1: Difference in received power when the distance or frequency is doubled (dB)

Carrier	0.5 → 1 km	1 → 2 km	4 → 8 km
900 MHz	6.0206	6.0206	6.0206
1900 MHz	6.0206	6.0206	6.0206
3500 MHz	6.0206	6.0206	6.0206
Doubling frequency 900 → 1800 MHz	6.0206 at d = 1 km	6.0206 at d = 3 km	—

Every case equals 20·log10(2) = 6.0206 dB, as the equation requires.

BER over AWGN

Table 2: BPSK sanity check: simulated against theoretical BER

Eb/N0 (dB)	Simulated	Theory 0.5·erfc(√(Eb/N0))
0	7.830e-02	7.865e-02
4	1.259e-02	1.250e-02
6	2.445e-03	2.388e-03
8	1.783e-04	1.909e-04
9	3.667e-05	3.363e-05

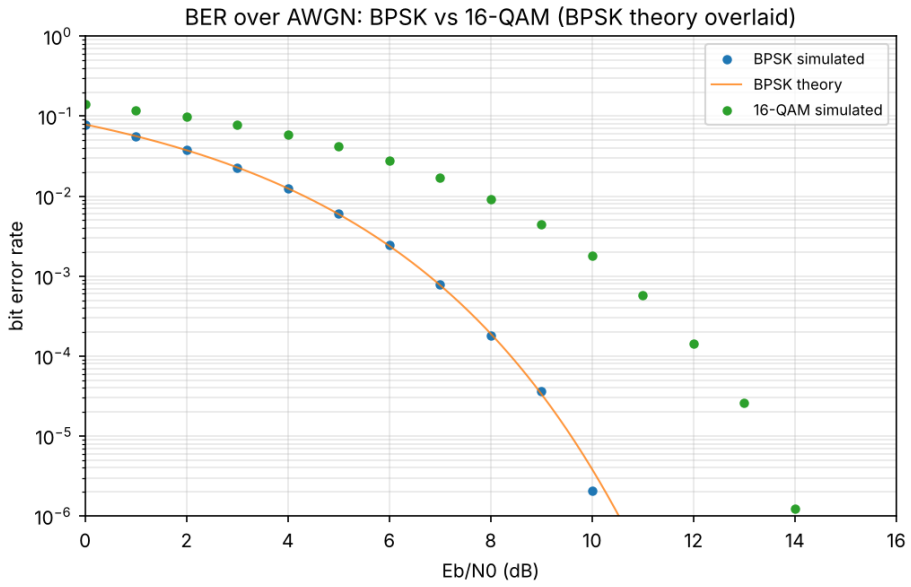


Figure 2: *Simulated BER of BPSK and 16-QAM with the theoretical BPSK curve overlaid.*

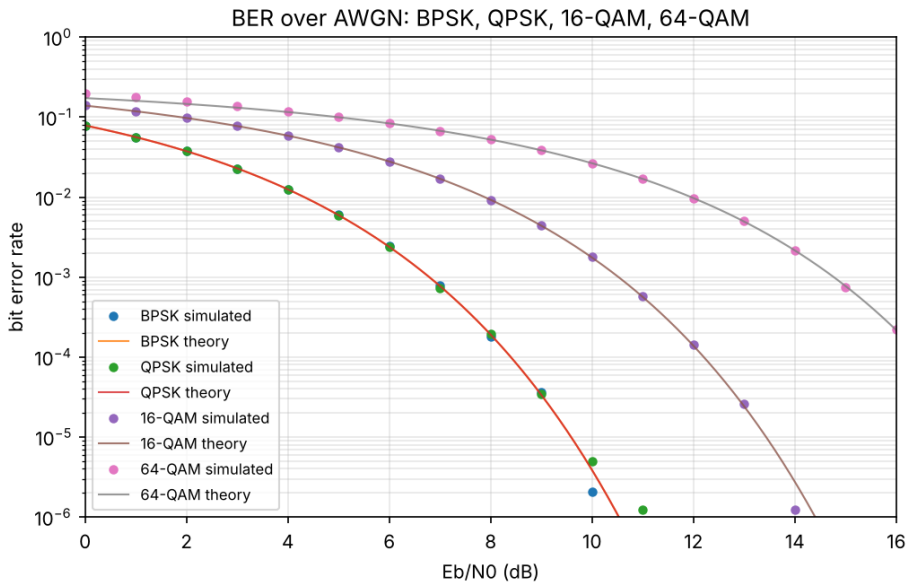


Figure 3: *Simulated BER (markers) and theory (lines) for BPSK, QPSK, 16-QAM and 64-QAM.*

The simulated BPSK points follow the theoretical curve closely, with the usual scatter at the lowest error rates, which is the sanity check required before trusting the other schemes. QPSK lies on the BPSK curve although it carries two bits per symbol. Each curve shows the classic waterfall shape, with the 16-QAM and 64-QAM waterfalls shifted to the right.

Table 3: Eb/N0 required for BER = 10^{-3} and distance from the Shannon limit (simulated)

Scheme	Bits per symbol	Eb/N0 (dB)	Es/N0 = SNR (dB)	Shannon minimum Eb/N0 (dB)	Distance above limit (dB)
BPSK	1	6.78	6.78	0.00	6.78
QPSK	2	6.74	9.75	1.76	4.98
16-QAM	4	10.52	16.54	5.74	4.78
64-QAM	6	14.73	22.51	10.21	4.52

The measured gap and its relation to Shannon-Hartley

At a target BER of 10^{-3}, 16-QAM needs **3.74 dB more Eb/N0** than BPSK, which is **9.76 dB more SNR (Es/N0)** because 16-QAM also carries four times as many bits per symbol (an extra 10·log10(4) = 6.02 dB in the conversion from Eb/N0 to Es/N0). QPSK needs 0.04 dB less Eb/N0 than BPSK, and 64-QAM needs 4.21 dB more than 16-QAM.

Note on the manual's expected observation. The manual states that 16-QAM needs roughly 8 to 10 dB more Eb/N0 than BPSK. Our measurement and the closed-form BER both give about 4 dB in Eb/N0. The 8 to 10 dB figure matches the SNR (Es/N0) gap of 9.76 dB, so the statement holds if it is read as an SNR gap.

The Shannon-Hartley theorem, C = B·log2(1 + SNR), implies that carrying η bits per second per hertz error-free requires SNR ≥ 2^η − 1. Going from BPSK (η = 1, SNR ≥ 0 dB) to 16-QAM (η = 4, SNR ≥ 15, or 11.76 dB) therefore costs at least 11.76 dB of SNR in the limit, and the uncoded schemes need 6.78 dB and 16.54 dB, a measured gap of 9.76 dB. The measured gap is smaller than 11.76 dB only because BPSK sits further above its limit (6.78 dB) than 16-QAM does (4.78 dB); neither scheme beats its own limit. The theorem does not predict the BER curve itself. It describes an ideal capacity-achieving code

with arbitrarily long blocks, whereas uncoded modulation with symbol-by-symbol minimum-distance decisions always stays several dB above the ceiling, and a denser constellation, with a smaller minimum distance for the same average energy, is hit harder by noise. This comparison assumes Nyquist signalling with one complex symbol per hertz.

Lab tasks

- **QPSK and 64-QAM (tasks 1 and 3):** QPSK needs 6.74 dB, within 0.04 dB of BPSK, so doubling the rate costs nothing in Eb/N0. 64-QAM needs 14.73 dB, 4.21 dB above 16-QAM, and the Eb/N0 requirement keeps growing as the log2(1 + SNR) term demands more SNR for each extra bit per symbol.
- **Rate-1/2 convolutional code with Viterbi decoding (task 2):** constraint length 3, generators 7 and 5 (octal), hard-decision Viterbi decoding, 1,000,000 information bits in 2000 blocks with two tail bits. The reported Eb/N0 is per information bit, so the noise is scaled by the code rate. Coding gain over uncoded BPSK is shown in Table 4 and Figure 4.
- **Log-distance model and shadowing (tasks 4 and 5):** $PL(d) = PL(d_o) + 10 \cdot n \cdot \log10(d/d_o)$ with d_o = 100 m, 1900 MHz, $PL(d_o)$ = 78.02 dB (free space), n = 2 to 4, and log-normal shadowing with σ = 6 dB on n = 3.5, six realizations (Figure 5). The shadowing samples are drawn independently at each distance, so they have no spatial correlation, which is a simplification.
- **Throughput proxy (task 6):** $k \cdot (1 - BER)$ per scheme against the Shannon curve log2(1 + SNR) (Figure 6). Every scheme stays below the capacity curve, and each saturates at its k bits per symbol once errors vanish, whereas capacity keeps growing logarithmically.

Table 4: Coding gain of the rate-1/2 convolutional code over uncoded BPSK

Target BER	Uncoded Eb/N0 (dB)	Coded Eb/N0 (dB)	Coding gain (dB)
10^{-3}	6.78	5.72	+1.06
10^{-4}	8.37	7.07	+1.29
10^{-5}	9.45	7.79	+1.67

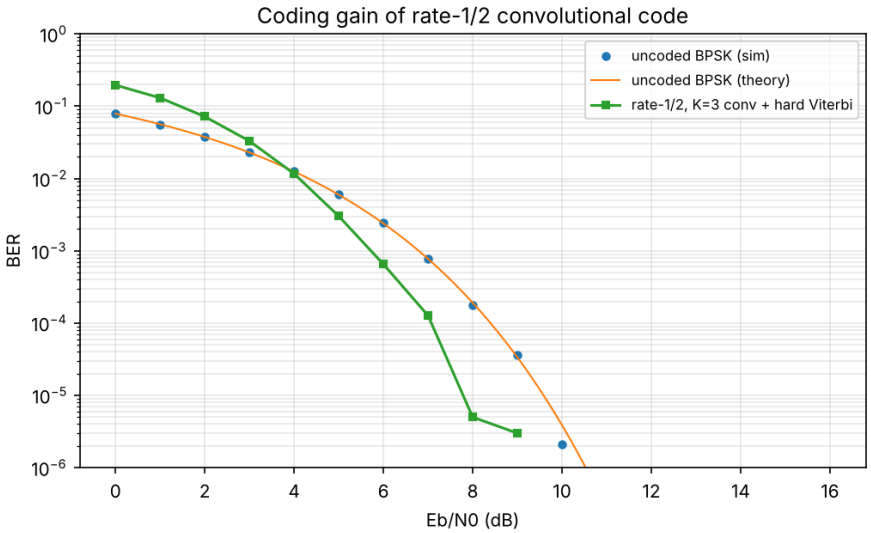


Figure 4: *Coded (rate-1/2, K = 3, hard Viterbi) against uncoded BPSK.*

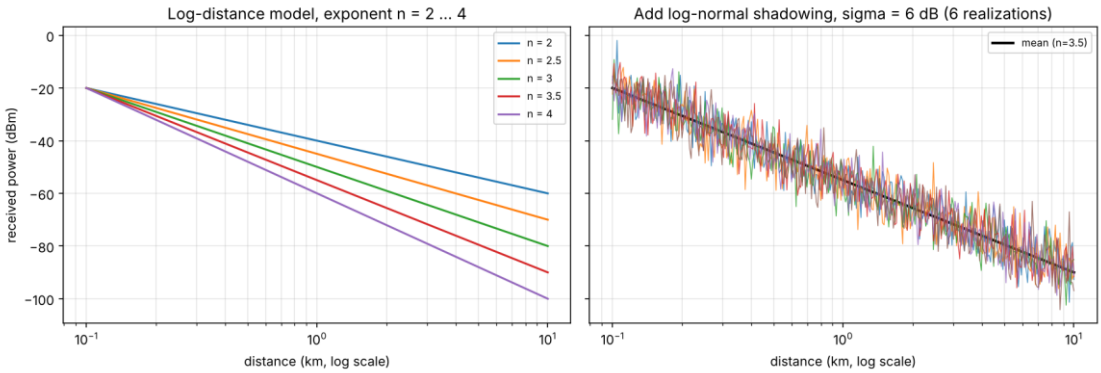


Figure 5: *Log-distance model for n = 2 to 4 (left) and six shadowing realizations with σ = 6 dB (right).*

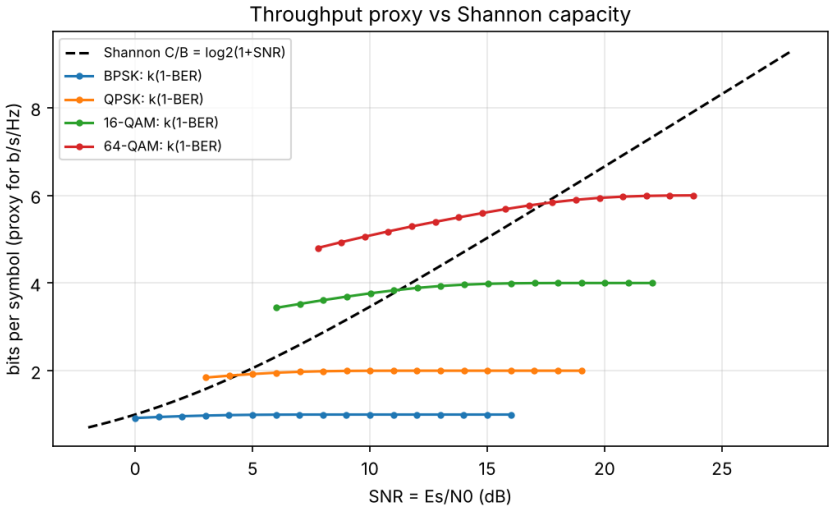


Figure 6: *Throughput proxy k·(1 − BER) against the Shannon capacity curve.*

3. Mini Project 2.2: OFDM Waveform Generator with a Cyclic-Prefix Visualizer

3.1 Objective

To build a functionally complete OFDM transmitter and receiver, send it through a multipath channel with a known delay spread, and show with measurable metrics that a cyclic prefix at least as long as the delay spread removes ISI, while a shorter or absent prefix does not.

3.2 System model

- **Transmitter:** random bits are mapped to Gray-coded QPSK or 16-QAM symbols, placed on N = 64 subcarriers, converted to the time domain by an IFFT, and a CP of P samples (a copy of the last P samples of the symbol) is prepended. Consecutive symbols are sent back to back so that smearing between adjacent symbols can occur.
- **Channel:** a three-tap FIR filter with taps at delays 0, 3 and 8 samples, gain magnitudes 0.864, 0.432 and 0.259 and phases 0, +0.8 and −1.9 rad (normalized to unit total power). The delay spread is therefore 8 samples, and the CP must be at least 8 samples long. The same channel realization is used for every receiver, with white Gaussian noise added for the SNR figures.

- **Receiver:** remove the CP, take the FFT, and equalize each subcarrier with one complex division by the known channel response (a one-tap zero-forcing equalizer). The weakest subcarrier has $|H| = 0.196$.
- **Metrics:** symbol error rate (SER), bit error rate and mean-squared error (MSE) between equalized and transmitted constellation points. The first symbol is excluded because it has no predecessor and so no ISI.

3.3 Results

Self-checks

Table 5: Verification of the building blocks

Check	Result
Mapper round trip, QPSK and 16-QAM (noiseless)	exact; average symbol energy 1
IFFT followed by FFT of the noiseless signal	maximum error 4.97×10^{-16}
CP of 16 samples against the symbol tail	identical
Noiseless, CP = 8 (equal to delay spread)	MSE 1.0×10^{-31}, SER 0 for QPSK and 16-QAM
Noiseless, CP = 0	QPSK: MSE 9.69×10^{-2}, SER 0.012; 16-QAM: MSE 8.71×10^{-2}, SER 0.097

The noiseless results isolate ISI: with a sufficient CP the receiver recovers the data to machine precision, and with no CP the errors come from the channel alone.

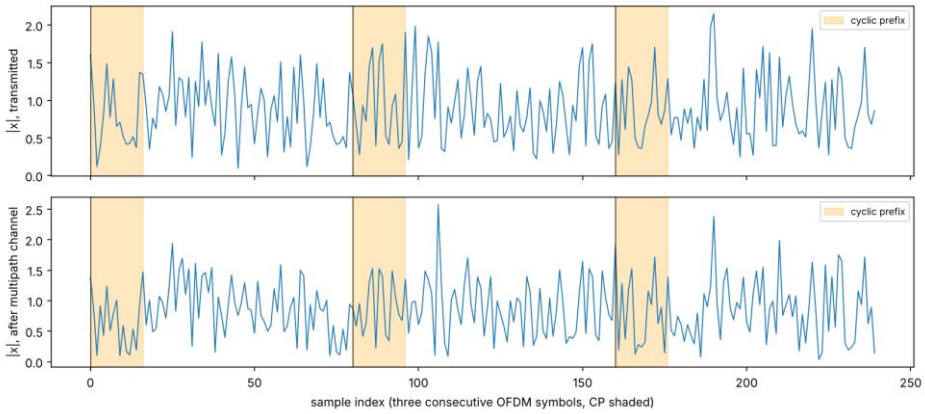


Figure 7: *Magnitude of three consecutive CP-OFDM symbols (CP shaded) before and after the multipath channel.*

Constellations with sufficient and insufficient CP

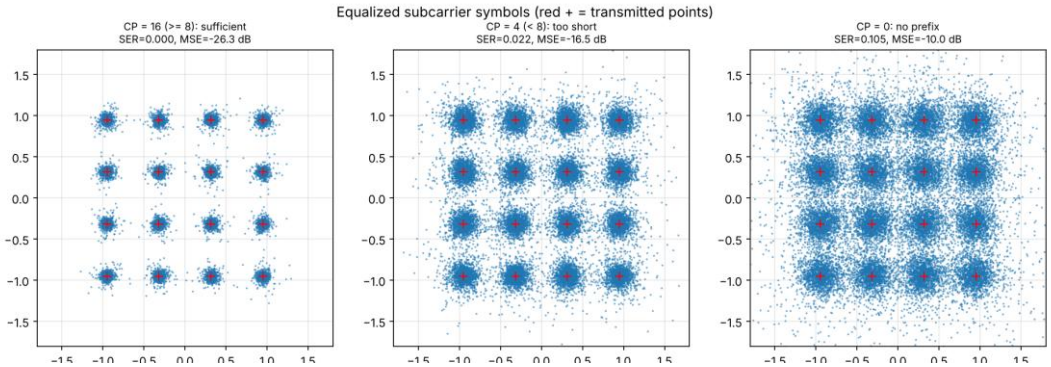


Figure 8: *Equalized 16-QAM symbols on the identical channel and data at 30 dB SNR: CP = 16, CP = 4 and CP = 0.*

Table 6: 16-QAM at 30 dB SNR on the same channel and data

CP length	SER	BER	MSE
16 (≥ 8, sufficient)	0.0002	0.0000	2.37×10^{-3} (−26.3 dB)
4 (< 8, too short)	0.0216	0.0058	2.26×10^{-2} (−16.5 dB)
0 (no prefix)	0.1054	0.0314	1.00×10^{-1} (−10.0 dB)

With a sufficient CP the received points form tight clusters around the transmitted points. With a CP of 4 the clusters visibly spread, and with no CP they smear into each other, which is the ISI the cyclic prefix is meant to prevent.

Error rate against CP length

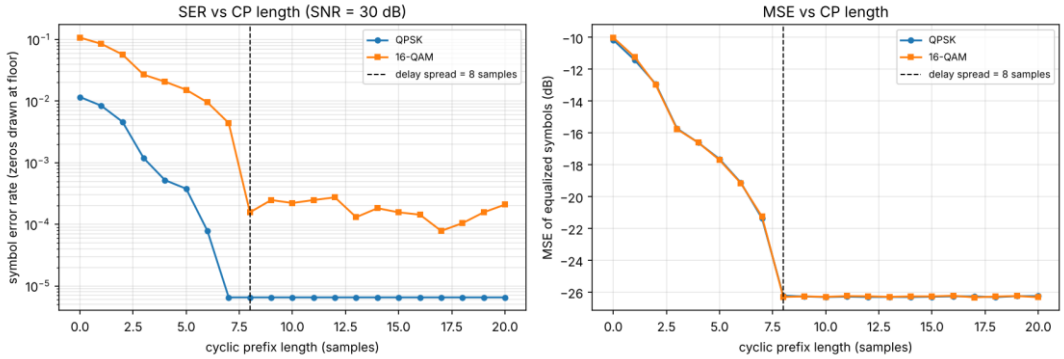


Figure 9: *SER (left) and MSE (right) against CP length at 30 dB SNR; the dashed line marks the 8-sample delay spread.*

Table 7: Error against CP length at 30 dB SNR (1200 symbols per scheme)

CP (samples)	QPSK SER	16-QAM SER	MSE (dB)	Useful fraction N/(N + P)
0	0.0116	0.1072	−10.0	100.0%
2	0.0046	0.0570	−13.0	97.0%
3	0.0012	0.0268	−15.8	95.5%
4	0.0005	0.0207	−16.6	94.1%
6	0.0001	0.0095	−19.2	91.4%
7	0	0.0044	−21.3	90.1%
8	0	0.0002	−26.3	88.9%
9	0	0.0002	−26.3	87.7%
12	0	0.0003	−26.3	84.2%
16	0	0.0001	−26.2	80.0%
20	0	0.0002	−26.3	76.2%

The MSE falls slowly from −10.0 dB at CP = 0, with a small step at CP = 3 (from −13.0 to −15.8 dB), where the delay-3 tap is fully absorbed. It improves gradually between CP = 4 and 7 as fewer samples of the delay-8 tap's echo remain in the FFT window, and then drops sharply from −21.3 dB at CP = 7 to −26.3 dB at CP = 8. From CP = 8 to 20 it stays flat at −26.3 dB (the MSE is the same for both modulations). QPSK shows no symbol errors from CP = 7 onward. 16-QAM settles at a noise-limited SER of roughly 1×10^{-4} to 3×10^{-4} from CP = 8 onward, because the equalizer amplifies noise on the weakest subcarrier.

3.4 Explanation

The cyclic prefix does two jobs. As a **guard interval**, it makes the multipath echoes of the previous symbol land inside the first P samples of the next one, and the receiver discards those samples, so no previous-symbol energy reaches the FFT window as long as P is at least the delay spread (8 samples here). As a **cyclic extension**, it is a copy of the symbol's own tail, so the channel's linear convolution with the transmitted block becomes a circular convolution over the N retained samples. The FFT then gives exactly $Y_k = H_k \cdot X_k$, which keeps the subcarriers orthogonal and lets a single complex division per subcarrier undo the channel; this is why the noiseless MSE is 10^{-31}.

When P is smaller than the delay spread, the last (8 − P) echo samples of the previous symbol leak into the FFT window, causing ISI, and the broken circularity also makes subcarriers interfere with each other. The error therefore falls as the CP grows, and drops sharply once P reaches 8. Beyond that point nothing more can be gained because only noise remains (the MSE is flat at −26.3 dB), yet every extra CP sample costs throughput: the useful fraction N/(N + P) is 88.9% at P = 8 but 76.2% at P = 20. The best CP is therefore just above the channel's delay spread.

4. Mini Project 2.3: Comparing WPAN, WLAN and WMAN Technologies

4.1 Objective

To capture the range, throughput and power trade-off across representative wireless personal-, local- and metropolitan-area technologies in a small sourced dataset, visualize it, and use it to recommend a technology for three deployment scenarios.

4.2 Dataset

The dataset covers two WPAN technologies (Bluetooth 5 and ZigBee), one WLAN (Wi-Fi 6, IEEE 802.11ax) and one WMAN (WiMAX, IEEE 802.16e). Range and rate figures are stated as spans because they depend heavily on environment, and the assumption behind each is recorded in `wireless_dataset.csv`. Source numbers refer to the reference list at the end of this section.

Table 8: Sourced specifications

Technology	Class	Range (m)	Peak data rate (Mbps)	Transmit power	Sources
Bluetooth 5 (BLE)	WPAN	40 (indoor, worst case) to 200 (outdoor line of sight)	0.125 (coded PHY) to 2 (2M PHY)	+4 dBm (LE Class 2); Class 1 up to +20 dBm	S1
ZigBee (IEEE 802.15.4)	WPAN	10 (nominal) to 100 (line of sight)	0.25 at 2.4 GHz	−3 dBm	S2
Wi-Fi 6 (802.11ax)	WLAN	under 50 (indoor) to 300 (outdoor)	4,803 (4-stream HE160 AP) to 9,600 (theoretical)	+28 dBm (5 GHz AP aggregate; EIRP limit 36 dBm)	S3, S4
WiMAX (802.16e)	WMAN	2,000 to 5,000 cell radius; fixed profile up to 10,000	15 (mobile peak) to 63 (downlink per sector); fixed profile 70	+30 dBm (base station, experimental link budget)	S5, S6, S7

Additional recorded facts: Bluetooth 5 application throughput is about 1.4 Mbps; ZigBee battery life is listed as 100 to 1,000+ days; the Wi-Fi 6 access point draws 26.1 W (single PoE port) to 32 W; permitted WiMAX EIRP reaches 61 dBm in the studied band.

Table 9: Derived quantities (peak rate divided by the stated quantity)

Technology	Rate per metre of range (Mbps/m)	Rate per mW of transmit power (Mbps/mW)	Rate × range (Mbps·km)
ZigBee	0.0025	0.50	0.025
Bluetooth 5	0.0100	0.80	0.400
Wi-Fi 6	32.0	15.2	2,880
WiMAX	0.0126	0.063	315

4.3 Trade-off visualization

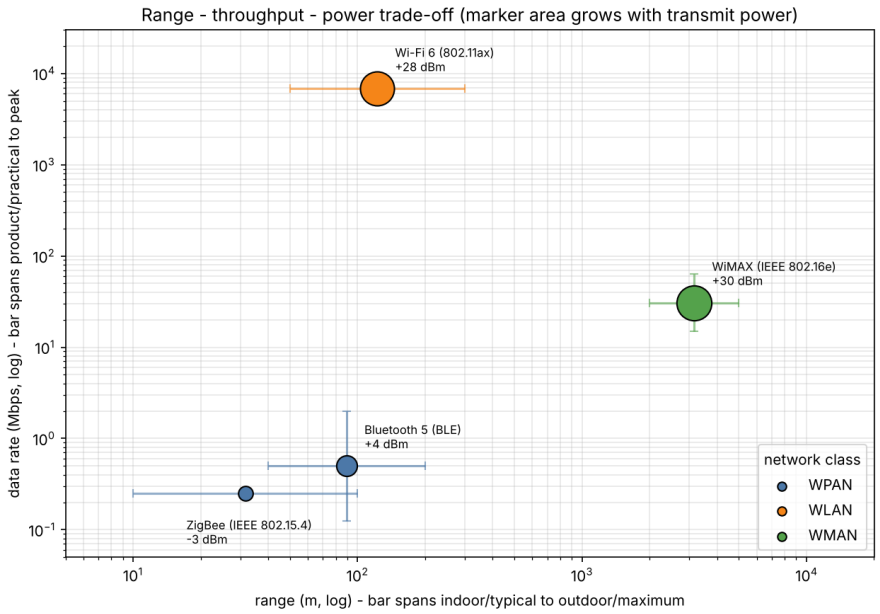


Figure 10: *Range against data rate on logarithmic axes; bars span the low-to-high end of each range, marker area grows with transmit power.*

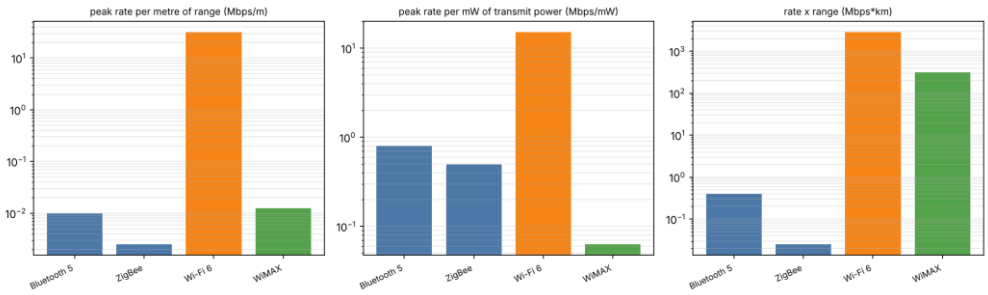


Figure 11: *Derived figures of merit on logarithmic axes: rate per metre of range, rate per mW and rate × range.*

The chart separates the technologies into three groups: the WPAN pair at low rate, short range and milliwatt power; Wi-Fi 6 at very high rate over a few hundred metres at high power; and WiMAX at moderate rate over kilometres, also at high power. Moving up the range axis from WPAN to WMAN buys reach, and moving up the rate axis buys throughput, but both cost transmit power, which is why no single technology dominates.

4.4 Deployment scenarios and recommendations

Scenario 1: Smart home

- R1 longest link about 25 m through walls
- R2 40 battery-powered sensors and switches
- R3 2 kbps per sensor, 0.08 Mbps in total
- R4 coin-cell power, so transmit power at most +10 dBm
- R5 battery life of at least 2 years

Recommendation: Bluetooth 5 (BLE). It is the only option that meets every numeric requirement: 40 m indoor range against 25 m needed, up to 2 Mbps against 0.08 Mbps, and +4 dBm (about 2.5 mW) against the +10 dBm ceiling. ZigBee has the lowest transmit power (−3 dBm, 0.5 mW) and a listed battery life of 100 to 1,000+ days, but its nominal 10 m single-hop range falls short of 25 m, so it would need mesh routers, and two years is reached only at the top of its quoted battery range. Wi-Fi 6 (+28 dBm, about 630 mW) and WiMAX (+30 dBm, 1,000 mW) exceed the power limit by more than 18 dB. Secondary factors: both WPAN technologies use unlicensed 2.4 GHz spectrum, and BLE can talk directly to phones without a hub (general knowledge, not from the sources). Caveat: no source gives BLE battery life in days. The one measured comparison in S1 (a 5-hour test) shows Bluetooth 5 using 31% (indoor) and 41% (outdoor) of its battery while the 802.15.4 node used 100%, but the set-ups differ, so it is indicative only.

Scenario 2: Campus network

- R1 0.49 km² campus (about 700 m by 700 m), indoors and outdoors
- R2 5,000 registered users, 20% active at once
- R3 1,000 active users × 2 Mbps = 2,000 Mbps aggregate
- R4 mains-powered infrastructure, no battery limit
- R5 at least 40 m per link indoors

Recommendation: Wi-Fi 6. Tiling the campus with 50 m cells (ideal, no overlap) needs 63 access points. At the product rate of 4,803 Mbps, derated to 10% for real shared-medium throughput (an assumption of this analysis), that is about 30,259 Mbps of aggregate capacity, roughly 15 times the 2,000 Mbps needed. WiMAX covers the whole campus from one base station but offers 15 to 63 Mbps per sector: even the 63 Mbps peak shared by 1,000 active users is 63 kbps each, 3.2% of the demand. BLE (98 cells, about 1.2 Mbps after derating) and ZigBee (10 m range) are not designed for this load. Secondary factors: Wi-Fi 6 uses unlicensed spectrum while WiMAX bands (2.3, 2.5 and 3.5 GHz) normally require a licence; 63 access points at 26.1 W is about 1.6 kW of PoE power plus cabling; S3 cites Target Wake Time and spatial reuse (up to a 30% throughput gain in dense outdoor deployments). Real deployments need more than 63 access points because circular cells never tile without overlap.

Scenario 3: Metro-area fixed link

- R1 one fixed 6 km link across a river, no fibre available
- R2 at least 20 Mbps sustained
- R3 mains or solar power, no battery constraint
- R4 no mobility, both ends fixed

Recommendation: WiMAX (802.16 fixed profile). It is the only technology whose range reaches 6 km: up to 10,000 m for the fixed profile (1.7 times the requirement) at up to 70 Mbps (3.5 times the 20 Mbps needed). The mobile 802.16e cell radius of 2 to 5 km would not reach 6 km, so the fixed configuration with a directional antenna is needed. Wi-Fi 6 offers the rate but only 300 m outdoors, 20 times too short; BLE (200 m) and ZigBee (100 m) fail on both range and rate. Caveats: WiMAX needs licensed spectrum in most countries; the experiment in S7 reached only about 300 m at 32 dBm EIRP and predicted about 2.1 km at 61 dBm EIRP, so range depends strongly on permitted power, antenna height and line of sight, and a 6 km link needs a proper link-budget study. WiMAX has largely been replaced by LTE and 5G in commercial networks (general knowledge, not from the sources), which affects equipment availability. Directional high-gain antennas can extend Wi-Fi links beyond 300 m, but no source in this dataset quantifies that.

Table 10: Summary of the requirement checks (computed by the script)

Technology	Scenario 1: smart home	Scenario 2: campus	Scenario 3: 6 km link
Bluetooth 5	meets all	fails aggregate (1.2 Mbps)	fails range and rate
ZigBee	fails range (10 m < 25 m)	fails range and aggregate (39 Mbps)	fails range and rate
Wi-Fi 6	fails power (+28 > +10 dBm)	meets all (about 30,259 Mbps)	fails range (300 m < 6,000 m)
WiMAX	fails power (+30 > +10 dBm)	fails aggregate (1.5 Mbps)	meets all (10,000 m, 70 Mbps)

4.5 Source notes and limits

- S1, S3 and S7 are academic papers; S2 is university course notes from about 2007, so the ZigBee figures are old; S4 is a vendor product page describing one Wi-Fi 6 access point; S5 and S6 are an educational website and lecture notes (secondary sources).
- The sources disagree on the WiMAX peak rate (15 Mbps mobile in S5, 63 Mbps downlink per sector in S6, 70 Mbps fixed in S5), so the table keeps the span instead of choosing one value.
- WiMAX subscriber-station power and Wi-Fi 6 client-device power were not found; the power column reflects access-point and base-station values. The “power class” labels in the data file and the 10% derating in Scenario 2 are the author's assumptions.

4.6 References

- S1: Bluetooth 5: a concrete step forward towards the IoT, arXiv:1711.00257, https://ar5iv.arxiv.org/html/1711.00257
- S2: IEEE ZigBee / 802.15.4 course notes, UNC Charlotte, https://webpages.charlotte.edu/~jmconrad/ECGR6185-2007-01/notes/IEEE_ZigBee.pdf
- S3: Revisiting Wireless Internet Connectivity: 5G vs Wi-Fi 6, arXiv:2010.11601, https://arxiv.org/pdf/2010.11601
- S4: Aruba AP-577 (802.11ax) product page, Secure Wireless Works, https://www.securewirelessworks.com/Aruba-AP-577.asp
- S5: Understanding IEEE 802.16 WiMAX standards, RF Wireless World, https://www.rfwireless-world.com/articles/understanding-ieee-802-16-wimax-standards
- S6: WiMAX lecture notes, McGill University, https://www.info413.ece.mcgill.ca/lecture%20note/E4%20WiMAX.pdf

- S7: WiMAX link budget and experiment, IEEE VTC Fall 2008, University of Bristol repository, https://research-information.bris.ac.uk/ws/files/3011100/tran_IEEE_VTC_fall2008.pdf

5. Conclusions

- The Friis equation behaves as predicted: every doubling of distance or frequency costs 6.0206 dB.
- The Monte Carlo BER curves match theory. At BER 10^{-3}, 16-QAM needs 3.74 dB more Eb/N0 (9.76 dB more SNR) than BPSK, and all schemes remain 4.5 to 6.8 dB above the Shannon limit; a rate-1/2 convolutional code recovers 1.06 to 1.67 dB of that.
- A cyclic prefix of at least the 8-sample delay spread cuts the 16-QAM symbol error rate at 30 dB from 10.5% (no prefix) to 0.02%, and longer prefixes only cost throughput.
- No single wireless technology wins on range, throughput and power together, so the recommendation depends on which requirement binds: power for the smart home, aggregate throughput for the campus, and range for the metropolitan link.